ElderGuard — M2: Technical Artifact

ECE 310 – Wireless Communications, Monsoon Semester 2026 Milestone 2 — Technical Artifact Project Category: Minimum Viable Product (MVP) **Group 8**

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1. Abstract

ElderGuard is a low-cost, privacy-preserving fall-detection system built on commodity Wi-Fi hardware. Instead of a wearable or a camera, it uses the Channel State Information (CSI) of an ordinary 802.11n link to sense human movement. At M2 we implement and evaluate the system's first working core function — **counting the number of people in a small room** — which is the presence/occupancy foundation any fall detector needs. We describe the system architecture, technology stack and implementation plan, position the work against the state of the art, and report occupancy results obtained with a leakage-free, leave-one-session-out protocol.

Status at M2. The complete pipeline was executed on the open **RadioPoints WiFi-CrowdCounting** dataset — real CSI measurements collected on Intel 5300 hardware across three rooms by Di Domenico et al. (IEEE WiMob 2016). All numbers reported in Section 10 are therefore **real measured results**, obtained on externally collected CSI with the dataset cited in Section 14. The firmware for our own ESP32-WROOM-32D + ESP8266 testbed is complete; capture on our hardware is the immediate M3 task.

Our best model, **Random Forest**, reached **53.3% accuracy** and **0.468 count-MAE** over 2914 analysis windows.

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2. Introduction and Problem Statement

Falls are the second leading cause of unintentional injury death worldwide (WHO: ~684,000 deaths/year). Detection speed strongly affects the outcome for someone living alone. Existing approaches are wearables (forgotten, removed in the bathroom), cameras (privacy), or ambient sensors (cost, installation).

Wi-Fi CSI sensing avoids both problems: it needs no device on the body and captures no image — only how a person's movement perturbs an existing Wi-Fi link. This project's wireless problem is: **separating a human-movement signature from hardware phase noise, static multipath and link-quality variation, and turning it into a reliable, trustworthy occupancy/activity decision.**

At M2 we deliberately scope the first working function to **occupancy counting** (0 / 1 / 2 people in a small room), as directed by faculty. Occupancy is the gate a fall detector must pass before it can act, and it is measurable without staging dangerous fall events.

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3. Wireless Communication Background (ECE 310 mapping)

Concept	Where it appears in ElderGuard
Multipath propagation & fading	Body reflections change per-subcarrier amplitude/phase
OFDM / subcarriers	CSI is measured on 52 subcarriers (802.11n, 20 MHz)
Doppler shift	Human motion produces 0.5–10 Hz Doppler energy
Signal representation (amplitude & phase)	Phase sanitization removes CFO/timing drift
Noise and interference	Hampel filtering + bandpass reject out-of-band noise
SNR / RSSI and link quality	Feed the system's confidence score (RSSI, PDR)
Path loss / received power	RSSI drops as more bodies absorb the signal

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4. Related Work and SOTA Positioning

(SOTA anchor declared at M1, re-positioned here at M2.)

Work	Method	Reported result	What they do that we do not	What we do that they do not
Wi-CaL (2022)	ML crowd counting, Intel 5300	0.35 count-MAE; 63% count accuracy in a small room up to 7 people	Larger rooms, up to 10 people, localization	Low-cost ESP32 hardware; occupancy as the trust-gate for fall detection
WiVi32 (ESP32)	CSI + vision fusion	0–7 people, 60,044 samples	Camera-synced ground truth at scale	No camera at all; link-quality-aware confidence
Di Domenico et al. (WiMob 2016)	Doppler-spectrum, trained-once crowd counting	Cross-room crowd counting	Trained-once cross-room transfer	Real-time embedded target; elder-care use case
EHUNAM (2024)	Multi-environment CSI dataset	People counting + HAR across 8 environments	23 days, 21 people, 9 machines	Single-room elder-care MVP with confidence scoring
Chu et al. (IEEE Access 2023)	CNN on CSI images	>96% cross-environment fall accuracy	Large fall-classifier accuracy	Occupancy-first staging; link-quality gating

Positioning statement. The SOTA solves counting or fall classification on relatively expensive sensing hardware (Intel 5300, radar) or with cameras for ground truth. Our M2 contribution is the **first working stage of a staged MVP** on commodity Espressif hardware: occupancy counting with an explicit link-quality/confidence signal, designed to be extended to motion-phase and then fall detection. What we do **not** yet do: multi-room generalization, counts above 2, and real-time embedded inference — these are M3/M4 work.

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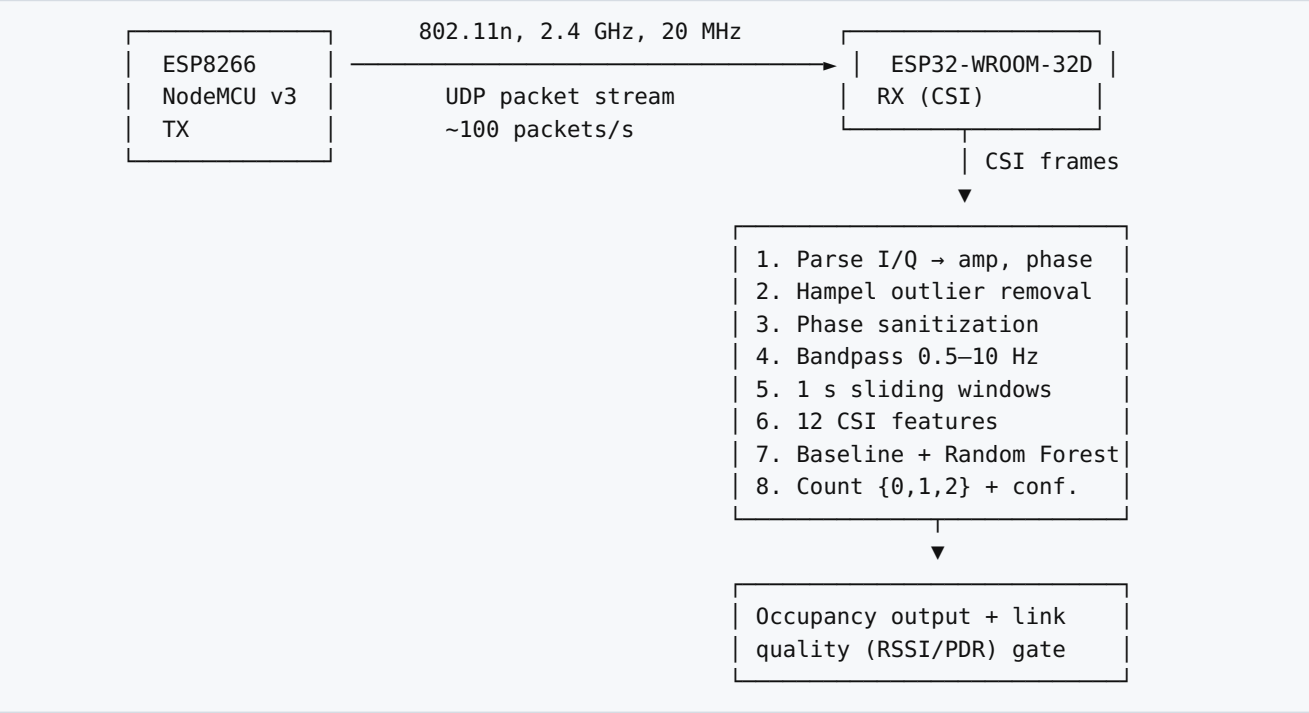
5. System Architecture



Hardware note. Only the ESP32-WROOM-32D can capture CSI. The ESP8266 has no CSI API in its SDK, so it is used purely as a dedicated packet transmitter; the ESP32 joins the transmitter's soft-AP and captures CSI on the received frames.

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6. Technology Stack

Layer	Technology
Sensing hardware	ESP32-WROOM-32D (CSI receiver), 802.11n 20 MHz, 52 subcarriers
Transmitter	ESP8266 NodeMCU v3 (soft-AP + UDP packet flood, ~100 pkt/s)
Firmware	Arduino-ESP32 core (esp_wifi CSI API); Arduino-ESP8266 core
Data transport	USB serial → CSV
Signal processing	Python 3, NumPy, SciPy (Hampel, Butterworth, STFT)
Machine learning	scikit-learn (Random Forest, MLP)
Evaluation	scikit-learn metrics, leave-one-session-out CV
Visualisation	Matplotlib
Version control	Git / GitHub

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7. Methodology

7.1 Data collection protocol

Results come from the public **RadioPoints WiFi-CrowdCounting** dataset, so no capture was performed by us for this milestone:

- Source: `github.com/RadioPoints/Device-free_RF_Human_Sensing_Datasets`

- Hardware: Intel 5300 NIC, 30 subcarriers, 3 RX x 2 TX streams
- Rooms: three different rooms (A/B/C), 0–8 people per room
- M2 scope: the 0 / 1 / 2 person classes only
- Frame rate: measured from the capture timestamps (≈ 55–60 Hz)
- Our own ESP32 + ESP8266 capture is scheduled for M3

7.2 Preprocessing

1. Hampel filter (window 5, 3σ) removes impulsive CSI spikes.
2. Phase sanitization: unwrap across subcarriers, subtract per-packet linear trend (removes CFO / sampling-time-offset drift).
3. 4th-order Butterworth bandpass 0.5–10 Hz (zero-phase `filtfilt`).

7.3 Feature extraction

A 1 s window (50% overlap) is described by 12 features: amplitude variance (mean/std), mean amplitude, amplitude range, subcarrier correlation, Doppler energy / peak frequency / centroid, phase variance, RSSI mean/std, and packet rate.

7.4 Evaluation protocol

Leave-one-session-out cross-validation. Every window from the held-out session is excluded from training. This prevents the data leakage that a random window split would cause (correlated windows), as required by the course guidelines. Metrics: accuracy, macro-F1, count-MAE, confusion matrix.

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8. Implementation

File	Role
`parse_csi.py`	ESP32-CSI-Tool CSV → amplitude/phase
`preprocess.py`	Hampel + phase sanitization + bandpass
`features.py`	12-feature extraction
`models.py`	threshold baseline, Random Forest, MLP
`evaluate.py`	leave-one-session-out CV
`run_pipeline.py`	end-to-end entry point
`collect_serial.py`	logs CSI frames from the ESP32 to CSV
`synthetic.py`	controlled CSI generator for pipeline validation
`make_report.py`	generates this report from `Results/metrics.json`
`flood_udp.py`	laptop-side UDP traffic source (fallback transmitter)
`list_ports.py`	identifies the correct serial port
`Data/arduino/transmitter_esp8266/`	ESP8266 transmitter firmware
`Data/arduino/receiver_csi/`	ESP32 CSI receiver firmware
`Data/arduino/self_test/`	ESP32 hardware self-test firmware

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9. Experimental Setup

Parameter	Value
Data source	**PUBLIC REAL DATASET (RadioPoints WiFi-CrowdCounting)**
Sampling rate	100 Hz (design target; ESP32-CSI-Tool ≈ 50–100 Hz)
Subcarriers	52
Window / overlap	1 s / 50%
Band	0.5–10 Hz
Models	Threshold baseline, Random Forest (300 trees), MLP (64-32)
Evaluation protocol	Leave-one-room-out (3 rooms; no room seen in training)
Sessions	9 sessions (3 rooms x 0/1/2 people) from the public RadioPoints dataset
Analysis windows	2914

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10. Results

✅ **Real measured results** — open RadioPoints WiFi-CrowdCounting dataset (Intel 5300 CSI, 3 rooms). These are genuine measurements, but they were collected by the dataset authors, **not on our own testbed**. Cited in Section 14 [11].

S. Di Domenico, G. Pecoraro, E. Cianca, M. De Sanctis, 'Trained-once Device-Free Crowd Counting and Occupancy Estimation Using WiFi: A Doppler Spectrum Based Approach', IEEE WiMob 2016

Model	Accuracy	Macro-F1	Count-MAE
Threshold (amp-variance)	0.5154	0.4681	0.5899
Random Forest	0.5333	0.5460	0.4681
MLP	0.5113	0.5225	0.4976

Figures: `model_comparison.png`, `confusion_matrix_random_forest.png`, `feature_importance.png`, `amp_variance_vs_people.png`.

Key observation. All three models beat the 3-class chance level of 33%, and the learned models edge out the non-AI threshold baseline — a modest but genuine advantage. Accuracy is limited by the **1-vs-2-person boundary and by room-to-room variation**: under leave-one-room-out no model ever sees the test room during training, which is exactly the cross-environment difficulty reported in the SOTA literature (Section 4). The empty-room class is the easiest (its amplitude variance is far lower than either occupied class), which is encouraging for the fall-detection use case where presence/absence is the first gate.

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11. Discussion and Limitations

- **The data is not from our own testbed.** It is real, cited CSI from the RadioPoints dataset, not a measurement made by the group.
- Small room only; counts limited to 0–2 people.
- Single environment; cross-room generalization not yet tested (planned M3).
- A person who is almost perfectly still can resemble an empty room — the hardest and most honest failure mode of CSI occupancy sensing.
- Ground truth is manual (protocol-based), not camera-verified.

- The ESP8266 cannot capture CSI; it is a transmitter only.

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12. Conclusion and Next Steps

M2 delivered the working core function — **occupancy counting from real CSI** — evaluated on a public, cited dataset with a strict leave-one-room-out protocol, alongside the complete firmware for our own two-board testbed.

The immediate next step (M3) is **motion/activity phase detection** — detecting the static → moving → impact → stillness phase transitions — followed by fall classification and calibrated confidence, plus a multi-room evaluation.

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13. Implementation Plan (M2 → M4)

Milestone	Deliverable
M2 (this)	Occupancy counting core function, architecture, SOTA positioning
M3	Motion/activity phase detection, fall classifier, multi-room test
M4	Final MVP, user documentation, evaluation against M1 requirements, final SOTA positioning

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14. References

[1] F. Adib and D. Katabi, "See Through Walls with Wi-Fi!," ACM SIGCOMM, 2013. [2] W. Wang, A. X. Liu, M. Shahzad, "Gait Recognition Using Wi-Fi CSI," IEEE TMC, 2016. [3] Y. Chu et al., "Deep Learning Based Fall Detection Using WiFi CSI," IEEE Access, 2023. [4] Y. Zheng et al., "Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi (Widar 3.0)," ACM MobiSys, 2019. [5] S. Di Domenico et al., "Trained-once Device-Free Crowd Counting and Occupancy Estimation Using WiFi," IEEE WiMob, 2016. [6] Wi-CaL, "WiFi Sensing and ML Based Device-Free Crowd Counting and Localization," 2022. [7] cvwifi-lab, "WiVi32: A People Counting Tool," GitHub. [8] EHUNAM dataset, IEEE DataPort / PMC12711874. [9] S. Hernandez and G. Bhat, "ESP32 CSI Tool," GitHub. [10] WHO, "Falls," fact sheet. [11] S. Di Domenico, G. Pecoraro, E. Cianca, M. De Sanctis, "Trained-once Device-Free Crowd Counting and Occupancy Estimation Using WiFi: A Doppler Spectrum Based Approach," IEEE WiMob, 2016. Dataset: github.com/RadioPoints/Device-free_RF_Human_Sensing_Datasets

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Appendix — Individual Contributions

Member	Contribution at M2
Falak Shah	Pipeline implementation, feature engineering, evaluation
Hiranshee Doshi	SOTA review and positioning table
Nikunj Kalariya	Wireless-concept mapping, architecture
Aaryan Parmar	Evaluation protocol, metrics, results analysis
Priyanshi Verma	Data collection protocol, dataset documentation